

VAPO

Value-Model-Based RL for Advanced Reasoning

60.4 on AIME 2024

Surpassing DAPO by 10+ points

ByteDance Seed | Qwen2.5-32B Base Model | No SFT Data

arXiv:2504.05118

Why Value Models?

Finer Credit Assignment for Long Chain-of-Thought Reasoning

Value-Model-Free (GRPO, DAPO)

- Trajectory-level advantage assignment
- Uniform credit across all tokens
- Final reward propagates equally
- Simple but imprecise for multi-step
- Subtle errors → catastrophic failures

Value-Model-Based (VAPO)

- Token-level credit assignment
- Trace each action's impact on returns
- Lower variance per token
- Well-trained model generalizes
- Better sample efficiency

Key Insight: For complex reasoning with subtle step-level errors, token-level precision in credit assignment is critical.

Challenge: Value models for long-CoT tasks face bias from bootstrapping, heterogeneous lengths, and reward sparsity.

Three Challenges

Plaguing Value-Model-Based RL in Long-CoT Tasks

1. Value Model Bias

Long trajectories amplify

bootstrapping errors

TD learning accumulates

errors over many steps

Initial value estimates

propagate and compound

2. Heterogeneous Lengths

Short vs long responses

need different λ in GAE

Short: lower λ = lower

variance (Monte Carlo)

Long: higher λ = lower

bias (TD learning)

3. Reward Sparsity

Only final `<eos>` token

receives reward signal

Long CoT = thousands of

tokens unguided

Gradient signal vanishes

for intermediate steps

VAPO Framework

Integrated Solutions for Value-Model-Based Reasoning RL

Value-Pretraining

Challenge 1: Bias

Pre-trains value model
to reduce initial bias

Decoupled-GAE

Challenge 1: Bias

Separates GAE computation
for robust estimation

Length-Adaptive GAE

Challenge 2: Heterogeneous
Lengths

Dynamic λ adjustment
based on response length

Clip-Higher

DAPO technique

Asymmetric clipping
encourages exploration

Token-level Loss

DAPO technique

Normalizes at token level
not sequence level

Positive Example LM Loss

Stability

Self-imitation learning
on correct trajectories

Group-Sampling

GRPO technique

Multiple responses
per prompt

Challenge Mapping:



Challenge 1: Bias from bootstrapping



Challenge 2: Heterogeneous sequence lengths



Challenge 3: Reward sparsity at length

Length-Adaptive GAE

Dynamic λ Adjustment for Variable Response Lengths

The Problem:

Standard GAE uses a fixed λ parameter that trades off bias vs variance:

- $\lambda \rightarrow 0$: Low variance, high bias (TD learning)
- $\lambda \rightarrow 1$: Low bias, high variance (Monte Carlo)

Length-Adaptive Solution:

VAPO dynamically adjusts λ based on response length:

- Short responses $\rightarrow$ Lower λ (reduce variance)
- Long responses $\rightarrow$ Higher λ (reduce bias)

This optimizes the bias-variance tradeoff across heterogeneous sequence lengths

Short Response (e.g., <2K tokens)

- Lower λ (e.g., 0.3-0.5)
- Favors Monte Carlo returns
- Lower variance is critical for short seqs
- bootstrapping errors less impactful

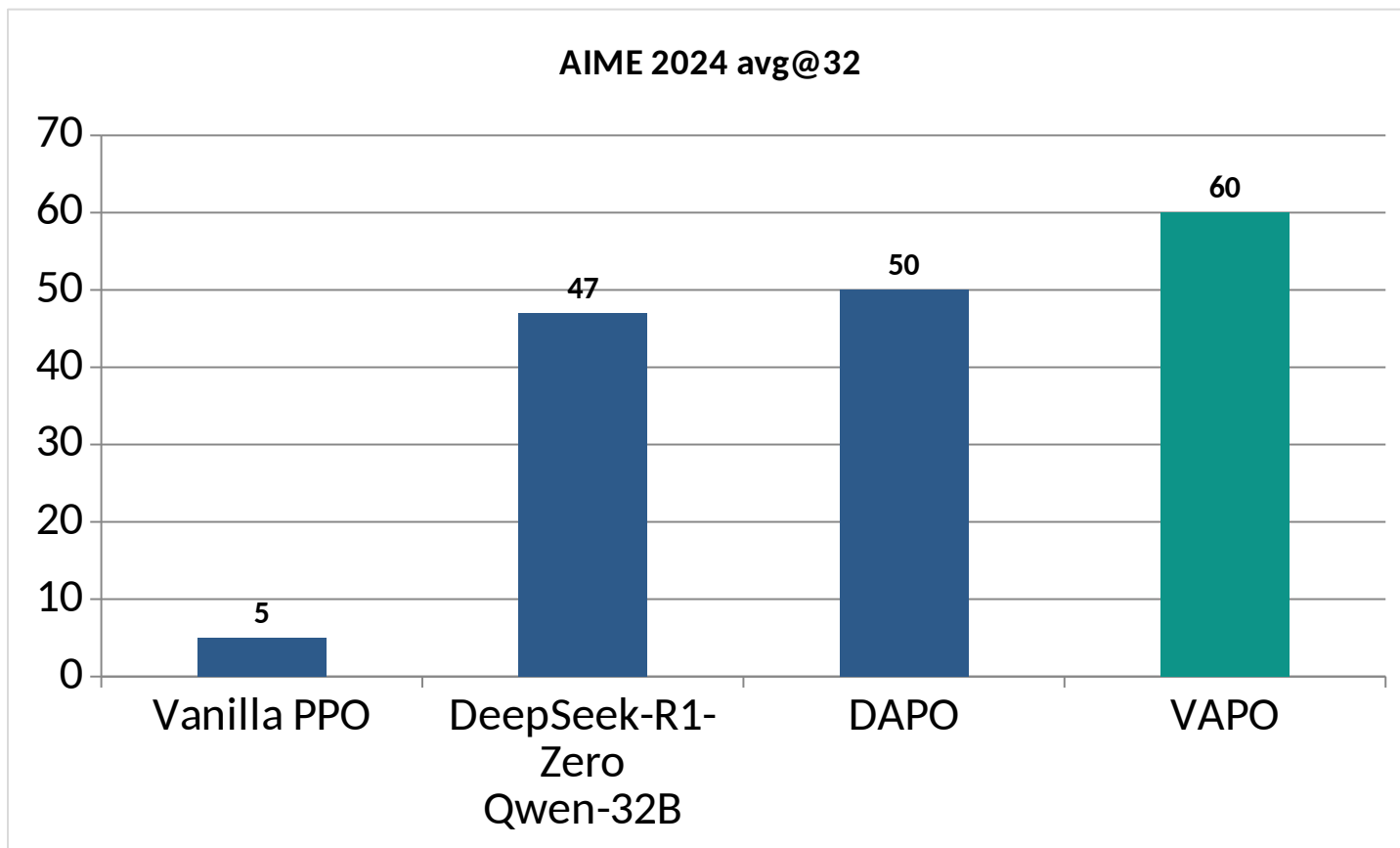
Long Response (e.g., >5K tokens)

- Higher λ (e.g., 0.7-0.9)
- Favors TD learning with bootstrapping
- Lower bias for long reasoning chains
- Accumulates less error over many steps

Contribution: Length-Adaptive GAE adds +15 points to AIME 2024 (45 $\rightarrow$ 60)

VAPO Scores 60 on AIME 2024

10+ Points Above DAPO and DeepSeek-R1-Zero



60.4 AIME 2024 avg@32

VAPO achieves SOTA

+10+ points over DAPO

Same base model: Qwen2.5-32B, no SFT

Training Efficiency

~5K steps vs DAPO's ~5.5K steps

See Figure 2: VAPO consistently outperforms DAPO throughout training, reaching 60% around step 4,500

Training Efficiency

VAPO Reaches SOTA in 5K Steps — Half the Training Budget of DAPO

Figure 2: AIME 2024 Accuracy vs Training Steps

- VAPO reaches 60% accuracy at ~step 4,500
- DAPO plateaus at ~50% at step 5,500
- DeepSeek-R1-Zero reaches 47% at ~10K steps
- VAPO consistently outperforms DAPO throughout entire training trajectory

Key Finding: VAPO achieves higher peak

Figure 3: Mean Response Length Over Training

- VAPO generates longer responses:
~6,000 tokens vs DAPO's ~4,500
- Longer responses suggest more elaborate reasoning chains
- VAPO's value model enables credit

Stability: VAPO shows zero training crashes across multiple independent runs

• DAPO's trajectory-level advantage

5K Steps to SOTA

Half DAPO's compute

60% Peak Accuracy

10+ pts above DAPO

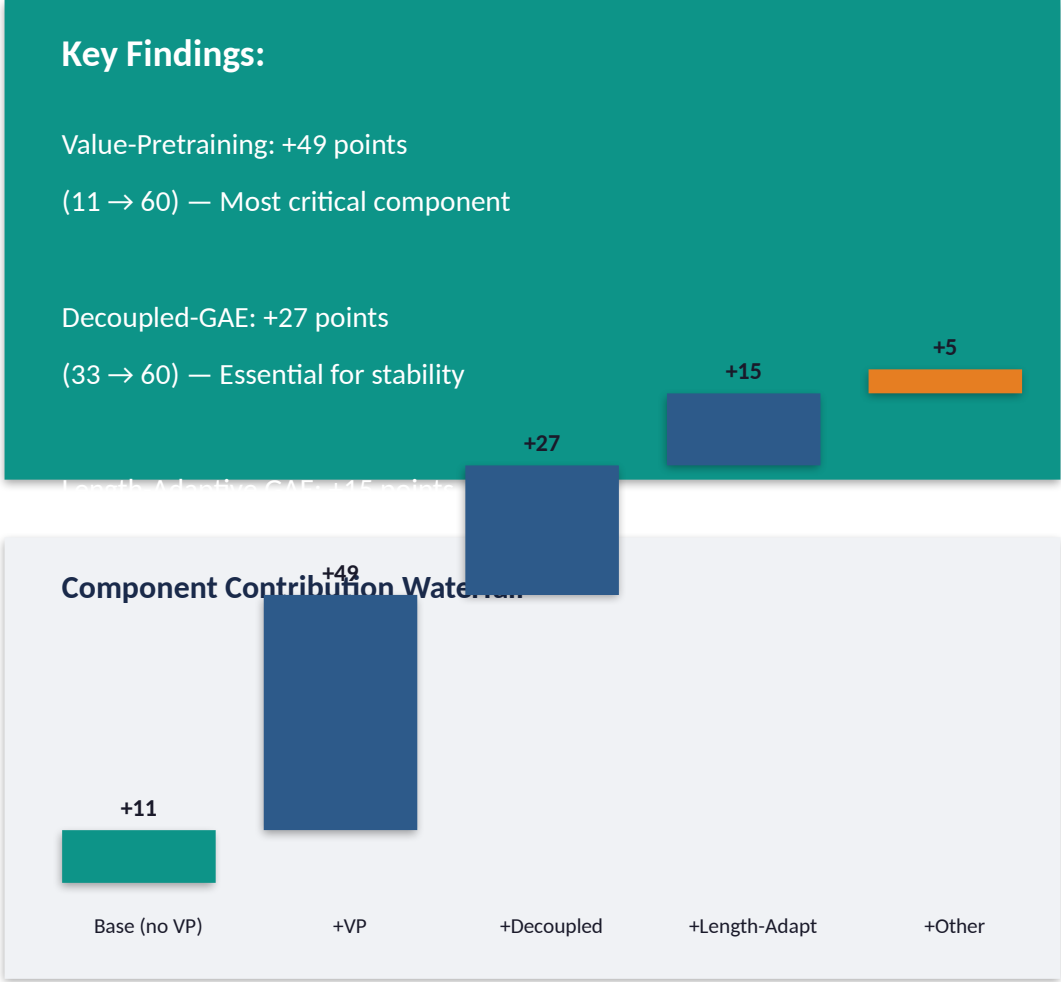
6K Avg Token Length

vs 4.5K for DAPO

Ablation Study

Every Component Matters — Value-Pretraining Alone Adds 49 Points

Configuration	AIME 2024 avg@32
VAPO (Full)	60
w/o Value-Pretraining	11
w/o Decoupled-GAE	33
w/o Length-adaptive GAE	45
w/o Clip-Higher	46
w/o Token-level Loss	53
w/o Positive Example LM Loss	54
w/o Group-Sampling	55



Key Takeaways

Value-Model-Based RL Is Back — With the Right Engineering

1

Value-Model-Based RL Can Surpass Value-Free Methods

When the three core challenges (bias, heterogeneous lengths, reward sparsity) are systematically addressed, value-model-based approaches can significantly outperform value-model-free methods on complex reasoning tasks.

2

Length-Adaptive GAE Is Critical for Long-CoT Tasks

Dynamic λ adjustment based on response length optimizes the bias-variance tradeoff across heterogeneous sequence lengths — a novel and effective technique unique to VAPO.

3

Value-Pretraining Is the Single Most Important Component

Contributing 49 points (11 $\rightarrow$ 60), value pretraining is indispensable. Without it, value-model-based RL barely works for long chain-of-thought reasoning.

4

VAPO Is Highly Stable with No Training Crashes

Reaching SOTA in $\sim$ 5,000 steps with zero crashes across multiple independent runs demonstrates the robustness of the VAPO framework for production deployment.

5

Implications for Future Reasoning Model Development

VAPO opens a promising direction: combining the credit assignment precision of value models with the stability techniques pioneered by DAPO/GRPO for next-generation reasoning systems.

VAPO: 60.4 on AIME 2024 — A new state-of-the-art for value-model-based reasoning RL